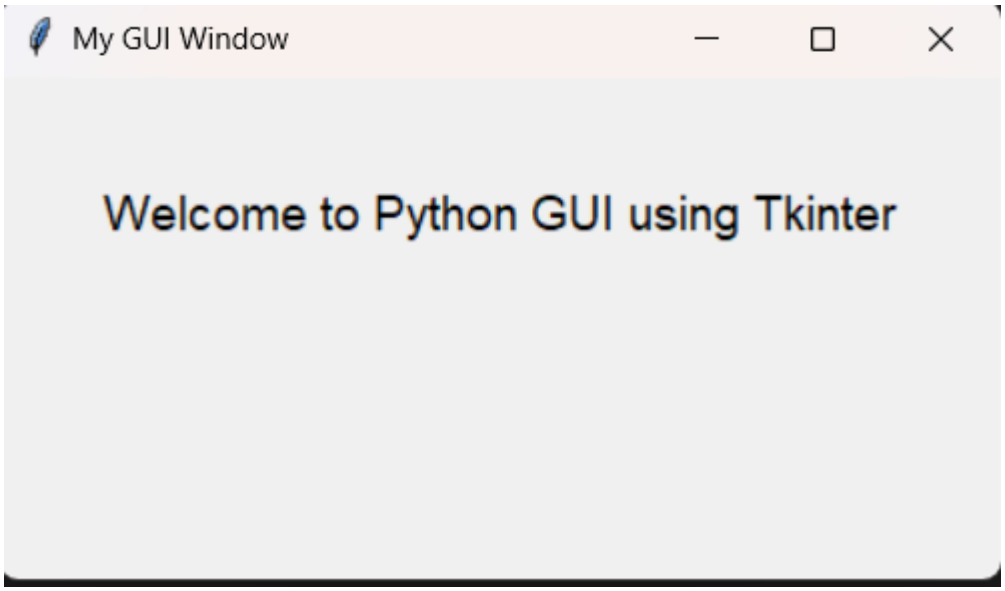
REDGNO : 24B01A1226

EXPERIMENT 1 : Write a Python GUI program to import Tkinter package and create a window. Set its title and add a label to the window.

CODE:

```
import tkinter as tk

window=tk.Tk()

window.title("My GUI Window")

window.geometry("400x200")

label=tk.Label(

    window,

    text="Welcome to Python GUI using Tkinter",

    font=("Arial",14)

)

label.pack(pady=40)

window.mainloop()
```

OUTPUT:



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EXPERIMENT 2: Write a Python program that designs a simple login form with labels and Entry widgets, arranging them in a grid using the Grid geometry manager.

CODE :

```
import tkinter as tk

root=tk.Tk()

root.title("Login Form")

root.geometry("300x200")

root.configure(bg="white")

lb_username=tk.Label(root,text="username :")

lb_username.grid(row=0,column=0,padx=10,pady=10,sticky="e")

entry_username=tk.Entry(root)

entry_username.grid(row=0,column=10,padx=10,pady=10)

lb_pass=tk.Label(root,text="Password :")

lb_pass.grid(row=1,column=0,padx=10,pady=10,sticky="e")

entry_pass=tk.Entry(root,show="*")

entry_pass.grid(row=1,column=10,padx=10,pady=10)

btn_login=tk.Button(root,text="Login")

btn_login.grid(row=3,column=0,columnspan=2,pady=15)

root.mainloop()
```

OUTPUT :

EXPERIMENT 3 : Write a program using BeautifulSoup4 library for web scraping for a given URL.

REDGNO : 24B01A1226
CODE :

```
import requests

from bs4 import BeautifulSoup

url="https://example.com/"

response=requests.get(url)

soup=BeautifulSoup(response.text,"html.parser")

if soup.title:

    print("Page Title:")

    print(soup.title.text)

else:

    print("No title found")

print("\nParagraphs:\n")

paragraphs = soup.find_all("p")

if paragraphs:

    for p in paragraphs:

        print(p.text)

else:

    print("No paragraphs found")
```

OUTPUT:

```
PS C:\Users\MOUNIKA\Desktop\django_rec_u1\project3> python web_scraping.py
Page Title:
Example Domain

Paragraphs:

This domain is for use in documentation examples without needing permission. Avoid use in operations.
Learn more
PS C:\Users\MOUNIKA\Desktop\django_rec_u1\project3>
```

EXPERIMENT 4: Develop a sample Hello World page using Flask framework.

CODE :

```
from flask import Flask
```

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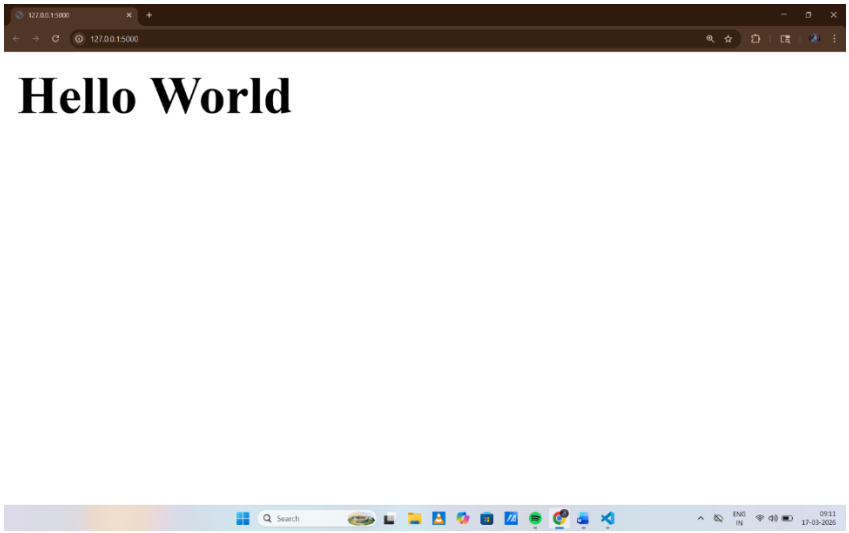
```
app=Flask(__name__)

@app.route("/")

def hello():

    return "<h1>Hello World</h1>"

if __name__=="__main__":

    app.run(debug=True)
```

OUTPUT:



EXPERIMENT 5:Develop a sample webpage

Using CherryPy :

CODE :

```
import cherrypy

class HelloWorld:

    @cherrypy.expose

    def index(self):

        return "<h1>Hello World from CherryPy!</h1>"

    @cherrypy.expose

    def about(self):
```

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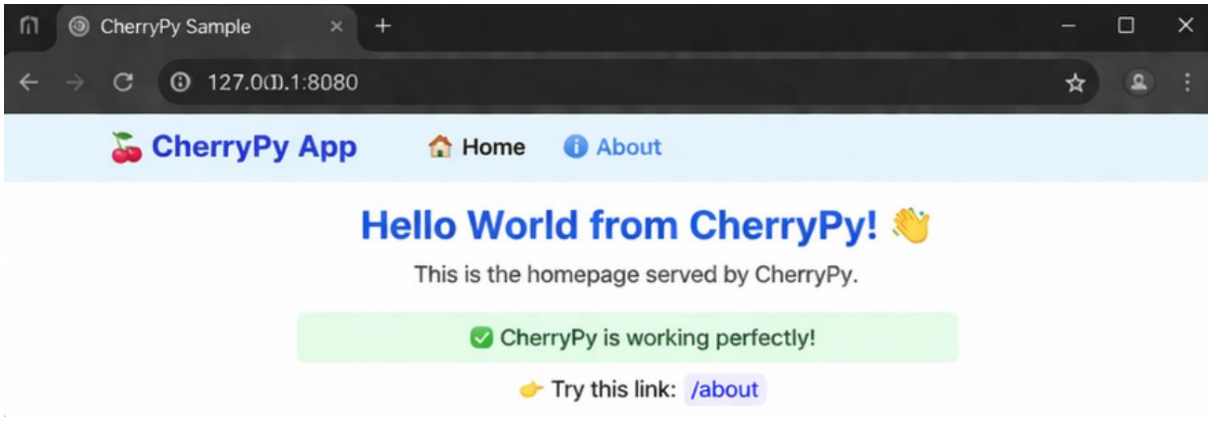
```
    return "<h2>This is a sample CherryPy webpage</h2>"

# Start server

if __name__ == '__main__':

  cherrypy.quickstart(HelloWorld())
```

OUTPUT :



Using Web2Py :

CODE :

```
web2py/
└── applications/
└── myapp/
 ├── controllers/
 │  └── default.py
 ├── views/
 │  └── default/
 │  └── index.html
```

applications/myapp/controllers/default.py :

```
def index():

  message = "Welcome to Web2Py Sample Webpage!"
```

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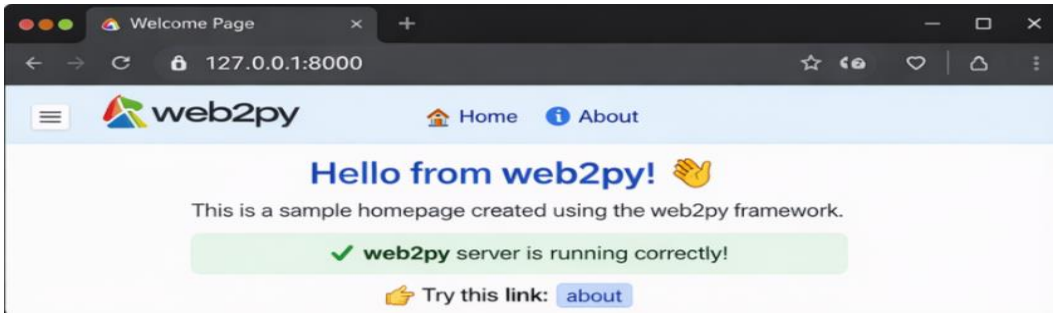
```
    return dict(message=message)
```

applications/myapp/views/default/index.html :

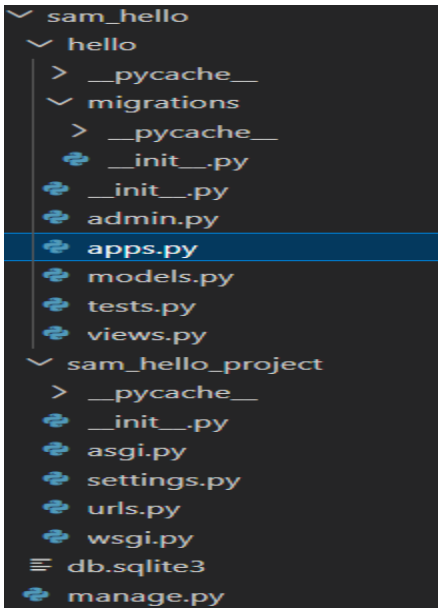
```
{{extend 'layout.html'}}

<h1>{{=message}}</h1>

<p>This is a simple webpage created using Web2Py.</p>

<button onclick="showMessage()">Click Me</button>

<script>

function showMessage() {

    alert("Hello! You clicked the button.");

}

</script>
```

OUTPUT :



EXPERIMENT 6: Create a Sample “Hello World” Application using Django

Folder structure :



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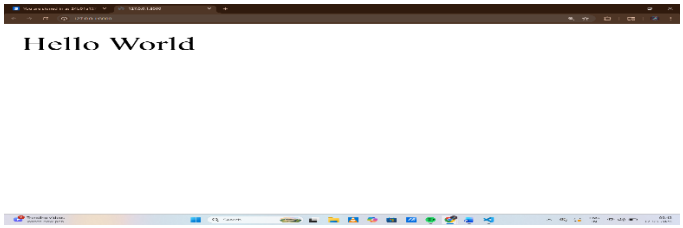
views.py :

```
from django.http import HttpResponse

def hello(request):

    return HttpResponse("Hello World from Django!")
```

urls.py :

```
from django.contrib import admin

from django.urls import path

from hello.views import hello

urlpatterns = [

    path('admin/', admin.site.urls),

    path('', hello),

]
```

OUTPUT :



EXPERIMENT 7 : Create a Login and Registration Page using MVC architecture in Django Framework

File structure:

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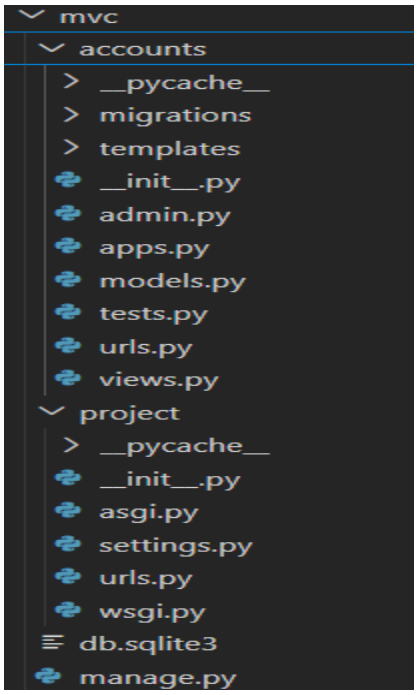

settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles'
     'accounts',
]
```

models.py :

```
from django.db import models
class User(models.Model):
    username = models.CharField(max_length=50
 email = models.EmailField()
    password = models.CharField(max_length=50)
    def __str__(self):
        return self.username
```

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```
python manage.py makemigrations
python manage.py migrate
```

views.py :

```
from django.shortcuts import render, redirect
from .models import Use
def register(request):
    if request.method == "POST":
        username = request.POST['username']
        email = request.POST['email']
        password = request.POST['password']
        User.objects.create(username=username, email=email, password=password)
        return redirect('login')
    return render(request, 'register.html')
def login(request):
    if request.method == "POST":
        username = request.POST['username']
        password = request.POST['password']
        user = User.objects.filter(username=username, password=password)
        if user:
            return render(request, 'success.html')
        else:
            return render(request, 'login.html', {'error': 'Invalid Credentials'})
    return render(request, 'login.html')
```

urls.py :

```
from django.urls import path
from . import views
urlpatterns = [
```

REDGNO : 24B01A1226

```
    path('register/', views.register, name='register'),

    path('login/', views.login, name='login'),

]
```

project/urls.py :

```
from django.contrib import admin

from django.urls import path, include

urlpatterns = [

    path('admin/', admin.site.urls),

    path('', include('accounts.urls')),

]
```

accounts/templates/register.html :

```
<h2>Register</h2>

<form method="post">

    {% csrf_token %}

    Username: <input type="text" name="username"><br><br>

    Email: <input type="email" name="email"><br><br>

    Password: <input type="password" name="password"><br><br>

    <button type="submit">Register</button>

</form>
```

login.html :

```
<h2>Login</h2

    Username: <input type="text" name="username"><br><br>

  Password: <input type="password" name="password"><br><br>

    <button type="submit">Login</button>
```

OUTPUT :

REDGNO : 24B01A1226

EXPERIMENT 8 :Create a sample page in Django by integrating BootStrap

CODE :

settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
```

REDGNO : 24B01A1226

```
    'django.contrib.contenttypes',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'home',
]
```

views.py :

```
from django.shortcuts import render
def index(request):
    return render(request, 'index.html')
```

home/urls.py :

```
from django.urls import path
from . import views
urlpatterns = [
    path('', views.index),
]
```

project/urls.py

```
from django.contrib import admin
from django.urls import path, include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('home.urls')),
]
```

home/templates/index.html :

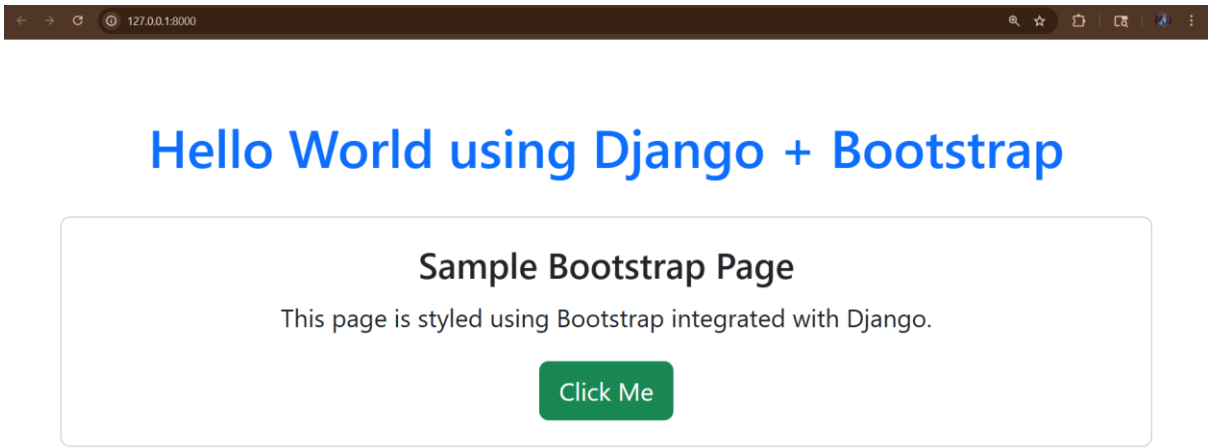
```
<!DOCTYPE html>
<html>
<head>
    <title>Bootstrap Django Page</title>
    <!-- Bootstrap CDN -->
```

REDGNO : 24B01A1226

```
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body>

<div class="container mt-5">

    <h1 class="text-primary text-center">

        Hello World using Django + Bootstrap

    </h1>

    <div class="card mt-4">

        <div class="card-body text-center">

            <h4 class="card-title">Sample Bootstrap Page</h4>

            <p class="card-text">

                This page is styled using Bootstrap integrated with Django.

            </p>

            <button class="btn btn-success">

                Click Me

            </button>

        </div>

    </div>

</div>

</body>

</html>
```

OUTPUT :

REDGNO : 24B01A1226



EXPERIMENT 9 : Create an application with Tables, grids in Djang

CODE :

products/models.py :

```
 from django.db import models
class Product(models.Model):
    name = models.CharField(max_length=50)
    price = models.IntegerField()
    category = models.CharField(max_length=50)
    def __str__(self):
        return self.name
```

python manage.py makemigrations

python manage.py migrate

REDGNO : 24B01A1226
products/views.py :

```
from django.shortcuts import render

from .models import Product

def show_products(request):

    products = Product.objects.all()

    return render(request, 'index.html', {'products': products})
```

products/urls.py :

```
from django.urls import path

from . import views

urlpatterns = [

    path('', views.show_products),

]
```

project/urls.py :

```
from django.contrib import admin

from django.urls import path, include

urlpatterns = [

    path('admin/', admin.site.urls),

    path('', include('products.urls')),

]
```

products/templates/index.html :

```
<!DOCTYPE html>

<html>

<head>

    <title>Table and Grid Example</title>

    <style>
```

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```
  table {
            border-collapse: collapse;
            width: 60%;
            margin-bottom: 40px;
        }
        table, th, td {
            border: 1px solid black;
            padding: 10px;

        .grid {
            display: grid;
            grid-template-columns: repeat(3, 1fr);
            gap: 10px;
        }
        .box {
            border: 1px solid black;
            padding: 20px;
            text-align: center;
        }
    </style>
</head>
<body>
<h2>Product Table</h2>
<table>
<tr>
<th>Name</th>
<th>Price</th>
<th>Category</th>
</tr>
```

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```
{% for p in products %}

<tr>

<td>{{ p.name }}</td>

<td>{{ p.price }}</td>

<td>{{ p.category }}</td>

</tr>

{% endfor %}

</table>

<h2>Product Grid</h2>

<div class="grid">

{% for p in products %}

<div class="box">

<h3>{{ p.name }}</h3>

<p>Price: {{ p.price }}</p>

<p>{{ p.category }}</p>

</div>

{% endfor %}

</div>

</body>

</html>
```

OUTPUT :

REDGNO : 24B01A1226

Product Grid

Laptop Price: 50000 Electronics	**Phone** Price: 20000 Electronics	**Chair** Price: 3000 Furniture

127.0.0.1:8000

Click to go back, hold to see history

Product Table

Name	Price	Category
Laptop	50000	Electronics
Phone	20000	Electronics
Chair	3000	Furniture

REDGNO : 24B01A1226

EXPERIMENT 10 : Create a Django App with Carousels feature

CODE :

Settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'collegeapp',
    'bootstrapapp',
]
```

Collegeproject/urls.py :

```
from django.contrib import admin
from django.urls import path, include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('collegeapp.urls')),
    path('bootstrap/', include('bootstrapapp.urls')),
]
```

App.py :

```
from django.apps import AppConfig
class CollegeappConfig(AppConfig):
    name = 'collegeapp'
```

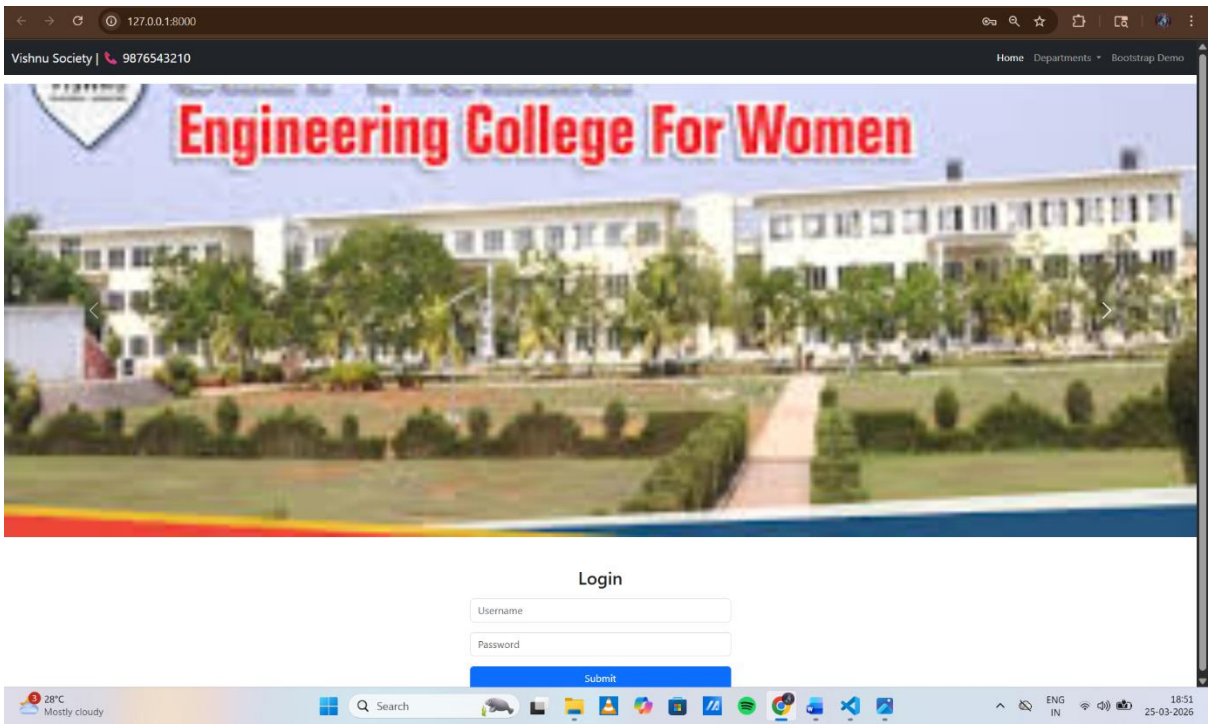
REDGNO : 24B01A1226
collegeapp/urls.py:

```
from django.urls import path

from . import views

urlpatterns = [

    path('', views.home, name='college_home'),

    path('department/<str:dept>/', views.department, name='department'),

]
```

Views.py:

```
from django.shortcuts import render

def home(request):

    return render(request, 'college_index.html')

def department(request, dept):

    dept = dept.upper()

    return render(request, 'department.html', {'dept': dept})
```

OUTPUT :



REDGNO : 24B01A1226

Welcome to CSE Department

This page belongs to the CSE branch.

Back to Home

REDGNO : 24B01A1226

EXPERIMENT 11 : Create a registration page using Authentication System

CODE :

Views.py :

```
from django.shortcuts import render, redirect
from django.contrib.auth.models import User
def register(request):
    if request.method == "POST":
        username = request.POST.get("username")
        email = request.POST.get("email")
        password = request.POST.get("password")
        User.objects.create_user(
            username=username,
            email=email,
            password=password
        )
        return redirect('/register')
    return render(request,"register.html")
```

urls.py :

```
from django.urls import path
from . import views
urlpatterns = [
    path('register/', views.register),
]
```

App.py :

```
from django.apps import AppConfig
class AccountsConfig(AppConfig):
    name = 'accounts'
```

REDGNO : 24B01A1226
settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'accounts',
]
```

Urls.py :

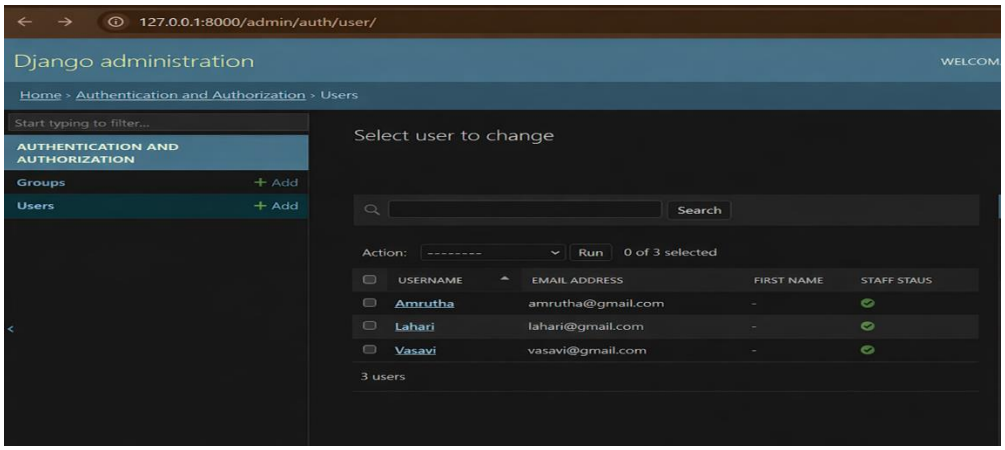
```
from django.contrib import admin
from django.urls import path,include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('accounts.urls')),
]
```

Manage.py

```
import os
import sys
def main():
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'project.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError from exc
    execute_from_command_line(sys.argv)
if __name__ == '__main__':   main()
```

OUTPUT :

REDGNO : 24B01A1226



EXPERIMENT 12 : Create an application in Django to send emails using email settings and Grid Layout

CODE :

Settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
```

REDGNO : 24B01A1226

```
    'users',
]
```

Urls.py :

```
from django.contrib import admin
from django.urls import path,include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('users.urls')),
]
```

Apps.py :

```
from django.apps import AppConfig
class UsersConfig(AppConfig):
    name = 'users'
```

urls.py:

```
from django.urls import path
from . import views
urlpatterns = [    path('', views.send_email), ]
```

views.py :

```
from django.core.mail import send_mail
from django.conf import settings
from django.http import HttpResponse
from django.shortcuts import render
def send_email(request):
    if request.method == "POST":
        email = request.POST.get("email")
        subject = request.POST.get("subject")
        message = request.POST.get("message")
        send_mail(
            settings.EMAIL_HOST_USER,
```

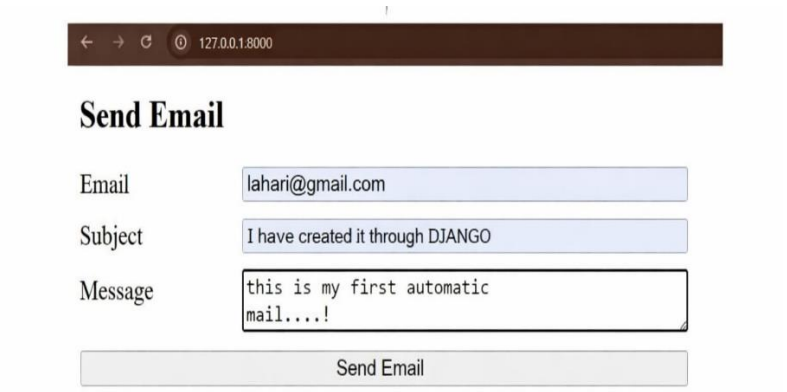
REDGNO : 24B01A1226

```
            [email],
            fail_silently=False,
        )
        return render(request, "success.html")
    return render(request, "email_form.html")
```

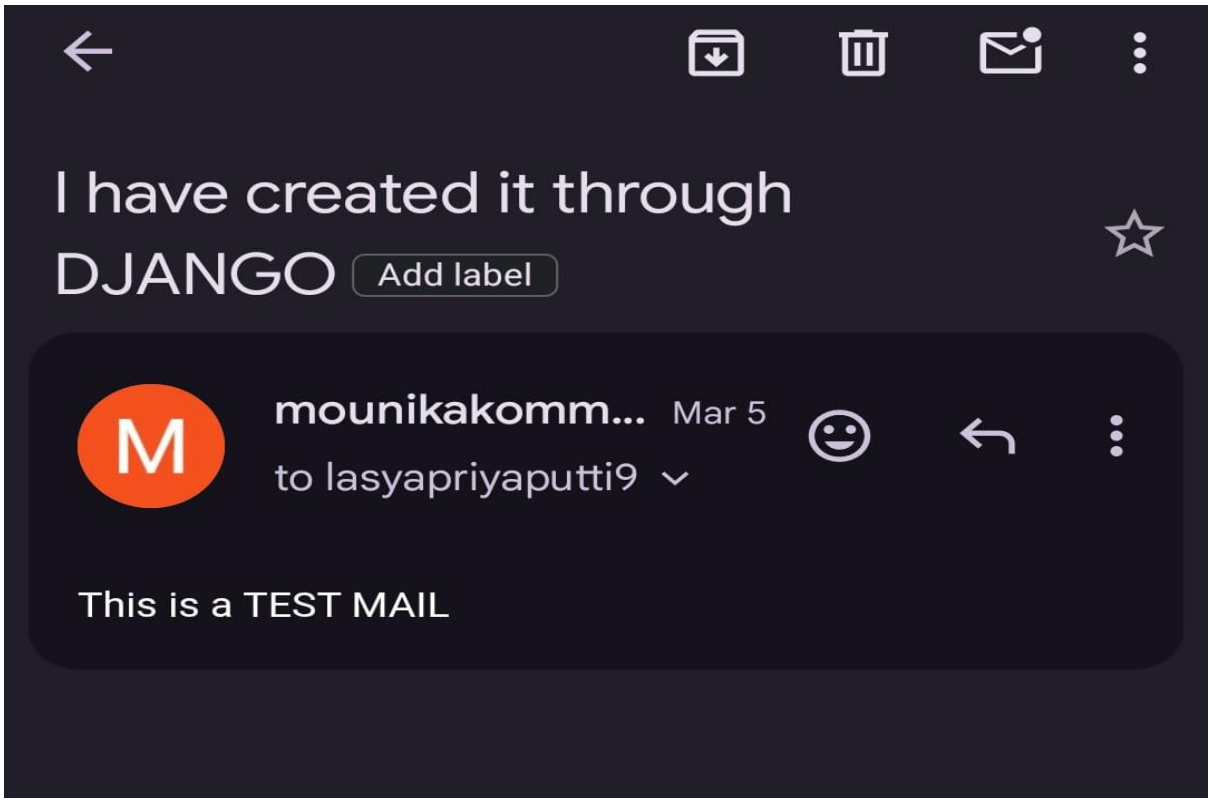
manage.py :

```
import os
import sys
def main()
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'myproject.settings')
   try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
    execute_from_command_line(sys.argv)
if __name__ == '__main__':     main()
```

OUTPUT :



REDGNO : 24B01A1226



EXPERIMENT 13 : Create an application in Django using page restriction /authentication with login and logut functionality.

CODE :

views.py :

```
from django.shortcuts import render

from django.contrib.auth.decorators import login_required

def login_view(request):

    return render(request, 'login.html')

@login_required

def home_view(request):

    render(request, 'accounts/home.html')
```

urls.py :

```
from django.contrib import admin

from django.urls import path, include

from django.contrib.auth import views as auth_views
```

REDGNO : 24B01A1226

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('accounts.urls')),
    path('accounts/login/',
auth_views.LoginView.as_view(template_name='accounts/login.html'), name='login'),
    path('logout/', auth_views.LogoutView.as_view(), name='logout'),
]
```

login.html :

```
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body><h2>Login Page</h2>
<form method="post">
    {% csrf_token %}
    {{ form.as_p }}
    <button type="submit">Login</button>
</form>
</body>
</html>
```

home.html :

```
<!DOCTYPE html>
<html>
<head>
    <title>Home</title>
</head>
<body>
<h2>Welcome {{ user.username }}</h2>
```

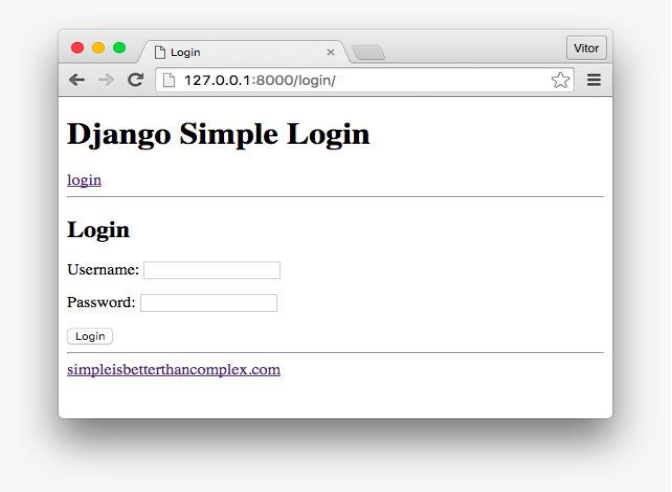
REDGNO : 24B01A1226

```
<p>This page is protected </p>

<a href="{% url 'logout' %}">Logout</a>

</body>

</html>
```

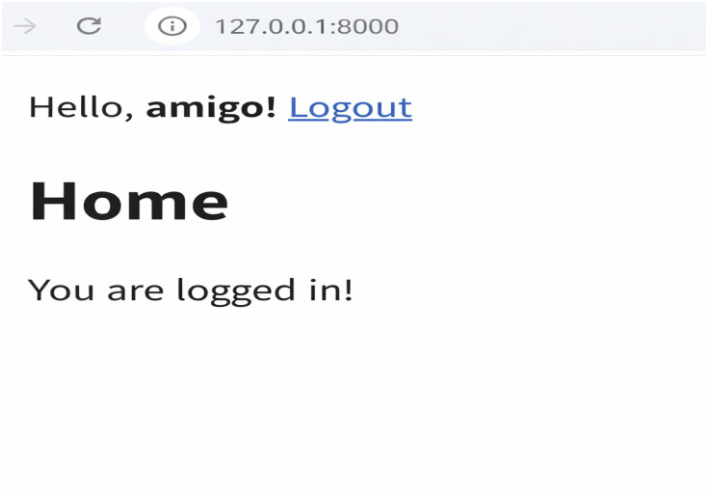
settings.py:

```
INSTALLED_APPS = [

    'django.contrib.admin',

    'django.contrib.auth',

    'django.contrib.contenttypes',

    'django.contrib.sessions',

    'accounts',

    'myproject',

]
```

OUTPUT :



REDGNO : 24B01A1226



EXPERIMENT 14 : Create a sample form using Django forms

CODE :

Settings.py :

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'formapp',
]
```

Urls.py :

```
from django.contrib import admin
from django.urls import path,include
urlpatterns = [
```

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```
    path('admin/', admin.site.urls),
    path('', include('formapp.urls')),
]
```

Views.py:

```
from django.shortcuts import render
from .forms import StudentForm
def student_form(request):
    if request.method == "POST":
        form = StudentForm(request.POST)
        if form.is_valid():
            form.save()
            return render(request,"success.html")
    else:
        form = StudentForm()
    return render(request,"student_form.html",{'form':form})
```

models.py :

```
from django.db import models
class Student(models.Model):
    name = models.CharField(max_length=100)
    email = models.EmailField()
    age = models.IntegerField()
    def __str__(self):
        return self.name
```

forms.py :

```
from django import forms
from .models import Student
class StudentForm(forms.ModelForm):
    class Meta:
        model = Student
```

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```
    fields = ['name','email','age']
```

app.py :

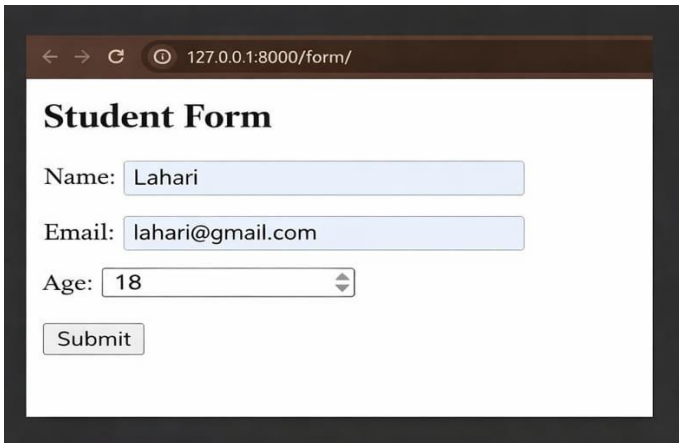
```
from django.apps import AppConfig

class FormappConfig(AppConfig):

    name = 'formapp'
```

manage.py :

```
import os

import sys

def main():

    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'project.settings')

    try:

        from django.core.management import execute_from_command_line

    except ImportError as exc:

        raise ImportError(

            "Couldn't import Django. Are you sure it's installed and "

            "available on your PYTHONPATH environment variable? Did you "

            "forget to activate a virtual environment?"

        ) from exc

    execute_from_command_line(sys.argv)

if __name__ == '__main__':

    main()
```

OUTPUT :

REDGNO : 24B01A1226



Form Submitted Successfully

Go Back

EXPERIMENT 15: Create an app in Django which fetches data from database and show as list and also save objects in database

CODE :

project/settings.py :

'items',

items/models.py :

```
from django.db import models

class Item(models.Model):

    name = models.CharField(max_length=100)

    quantity = models.IntegerField()

    def __str__(self):

        return self.name
```

python manage.py makemigrations

python manage.py migrate

items/admin.py

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```
from django.contrib import admin

from .models import Item

admin.site.register(Item)
```

items/views.py :

```
from django.shortcuts import render

from .models import Item

def item_list(request):

    items = Item.objects.all()

    return render(request, 'index.html', {'items': items})
```

items/urls.py :

```
from django.urls import path

from . import views

urlpatterns = [

    path('', views.item_list),]
```

project/urls.py :

```
from django.contrib import admin

from django.urls import path, include

urlpatterns = [

    path('admin/', admin.site.urls),

    path('', include('items.urls')),

]
```

items/templates/ :

items/templates/index.html :

```
<!DOCTYPE html>

<html>

<head>

    <title>Item List</title>

</head>

<body>
```

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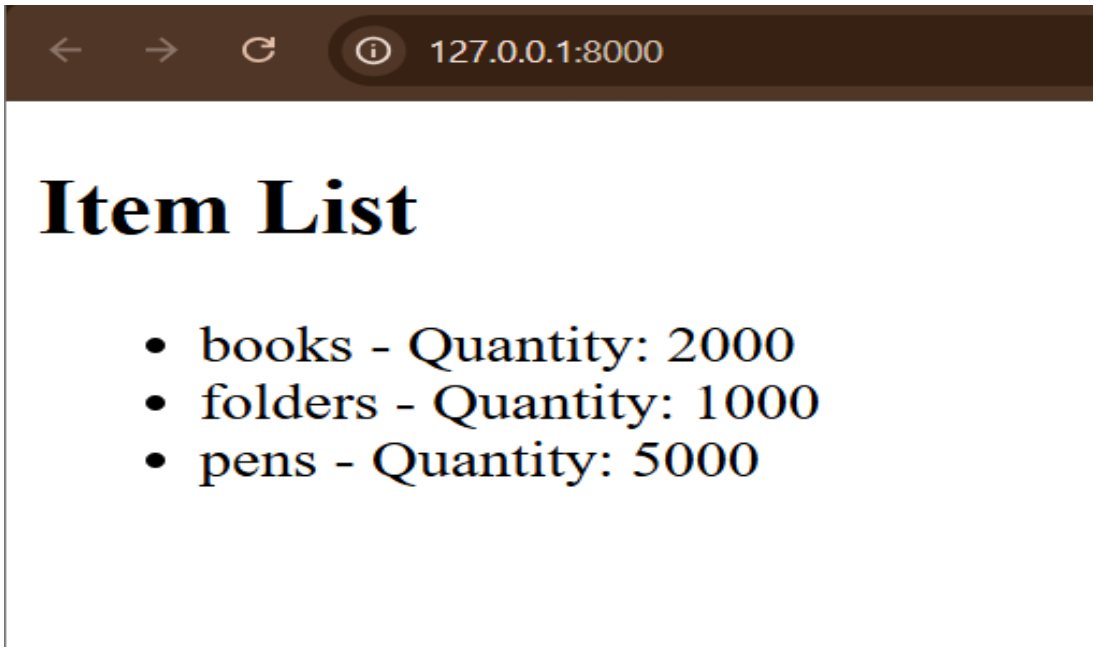
```
<h2>Item List</h2>

<ul>

{% for item in items %}

<li>{{ item.name }} - Quantity: {{ item.quantity }}</li>

{% endfor %}

</ul>

</body>

</html>
```

manage.py:

```
import os

import sys

def main():

    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'project.settings')

    try:

        from django.core.management import execute_from_command_line

    except ImportError as exc:

        raise ImportError(

            "Couldn't import Django. Are you sure it's installed and "

            "available on your PYTHONPATH environment variable? Did you "

            "forget to activate a virtual environment?"

        ) from exc

    execute_from_command_line(sys.argv)

if __name__ == '__main__':

    main()
```

OUTPUT :

REDGNO : 24B01A1226



EXPERIMENT 16 : create an app in Django for performing crud operations on records in a database

project/settings.py

```
INSTALLED_APPS = [
    'student',
]
```

student/models.py :

```
from django.db import models
class Student(models.Model):
    name = models.CharField(max_length=100)
    age = models.IntegerField()
    course = models.CharField(max_length=100)
    def __str__(self):
        return self.name
```

python manage.py makemigrations
python manage.py migrate

student/views.py :

```
from django.shortcuts import render, redirect
```

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```
from .models import Student

def index(request):

    students = Student.objects.all()

    if request.method == "POST":

        name = request.POST['name']

        age = request.POST['age']

        course = request.POST['course']

        Student.objects.create(name=name, age=age, course=course)

        return redirect('/')

    return render(request, 'index.html', {'students': students})

def update(request, id):

    student = Student.objects.get(id=id)

    if request.method == "POST":

        student.name = request.POST['name']

        student.age = request.POST['age']

        student.course = request.POST['course']

        student.save()

        return redirect('/')

    return render(request, 'update.html', {'student': student})

def delete(request, id):

    student = Student.objects.get(id=id)

    student.delete()

    return redirect('/')
```

student/urls.py :

```
from django.urls import path

from . import views

urlpatterns = [

    path('', views.index),

    path('update/<int:id>/', views.update),
```

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```
    path('delete/<int:id>/', views.delete),
]
```

project/urls.py :

```
from django.contrib import admin
from django.urls import path, include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('student.urls')),
]
```

student/templates/:

```
<h2>Add Student</h2>
<form method="post">
    {% csrf_token %}
    Name: <input type="text" name="name"><br><br>
    Age: <input type="number" name="age"><br><br>
    Course: <input type="text" name="course"><br><br>
    <button type="submit">Add</button>
</form>
<h2>Student List</h2>
<table border="1">
<th>Name</th>
<th>Age</th>
<th>Course</th>
<th>Actions</th>
{% for s in students %}
<td>{{ s.name }}</td>
<td>{{ s.age }}</td>
<td>{{ s.course }}</td>
```

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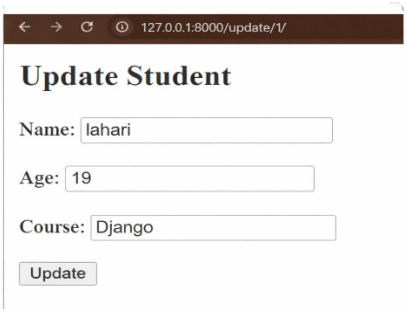
```
    <a href="update/{{ s.id }}">Edit</a> |

    <a href="delete/{{ s.id }}">Delete</a>

{% endfor %}

</table>
```

update.html (Update) :

```
<h2>Update Student</h2>

<form method="post">

    {% csrf_token %}

    Name: <input type="text" name="name" value="{{ student.name }}"><br><br>

    Age: <input type="number" name="age" value="{{ student.age }}"><br><br>

    Course: <input type="text" name="course" value="{{ student.course }}"><br><br>

    <button type="submit">Update</button>

</form
```

OUTPUT :

REDGNO : 24B01A1226



EXPERIMENT 17 : Create a website on Django with a login and registration page

CODE:

myproject/settings.py:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'myapp',   # ADD THIS
]
```

myproject/urls.py:

```
from django.contrib import admin
from django.urls import path
from myapp import views
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', views.home, name='home'),
    path('register/', views.register, name='register'),
    path('login/', views.user_login, name='login'),
    path('logout/', views.user_logout, name='logout'),
```

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]

myapp/views.py:

```
from django.shortcuts import render, redirect
from django.contrib.auth.models import User
from django.contrib.auth import authenticate, login, logout


def home(request):
    return render(request, 'home.html')
def register(request):
    if request.method == 'POST':
        username = request.POST['username']
        password = request.POST['password']
        User.objects.create_user(username=username, password=password)
        return redirect('login')
    return render(request, 'register.html')
def user_login(request):
    if request.method == 'POST':
        username = request.POST['username']
        password = request.POST['password']
        user = authenticate(request, username=username, password=password)
        if user:
            login(request, user)
            return redirect('home')
    return render(request, 'login.html')
def user_logout(request):
    logout(request)
    return redirect('login')
```

templates/home.html:

```
<h1>Welcome Home Page</h1>
```

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```
{% if user.is_authenticated %}
    <p>Hello, {{ user.username }}</p>
    <a href="/logout/">Logout</a>
{% else %}
    <a href="/login/">Login</a>
    <a href="/register/">Register</a>
{% endif %}
```

templates/register.html:

```
<h2>Register</h2>
<form method="POST">
    {% csrf_token %}
    Username: <input type="text" name="username"><br><br>
    Password: <input type="password" name="password"><br><br>
    <button type="submit">Register</button>
</form>
<a href="/login/">Already have an account? Login</a>
```

templates/login.html:

```
<h2>Login</h2>
<form method="POST">
    {% csrf_token %}
    Username: <input type="text" name="username"><br><br>
    Password: <input type="password" name="password"><br><br>
    <button type="submit">Login</button>
</form>
<a href="/register/">Create account</a>
```

python manage.py makemigrations
python manage.py migrate

OUTPUT :

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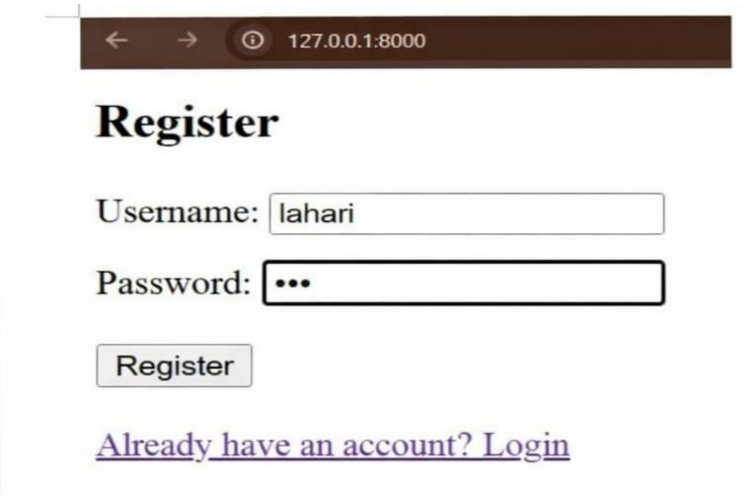

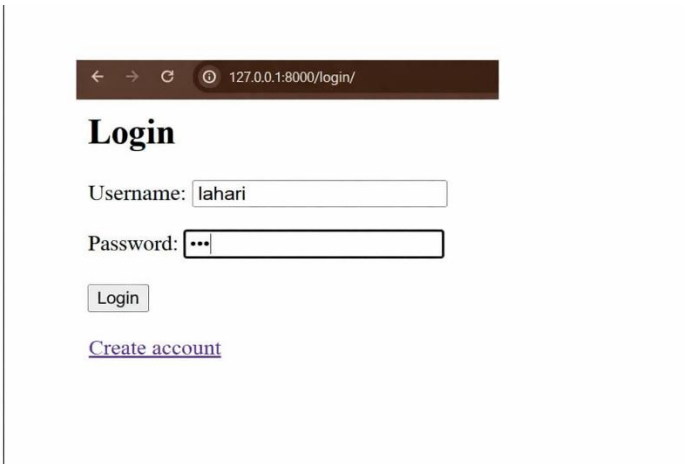

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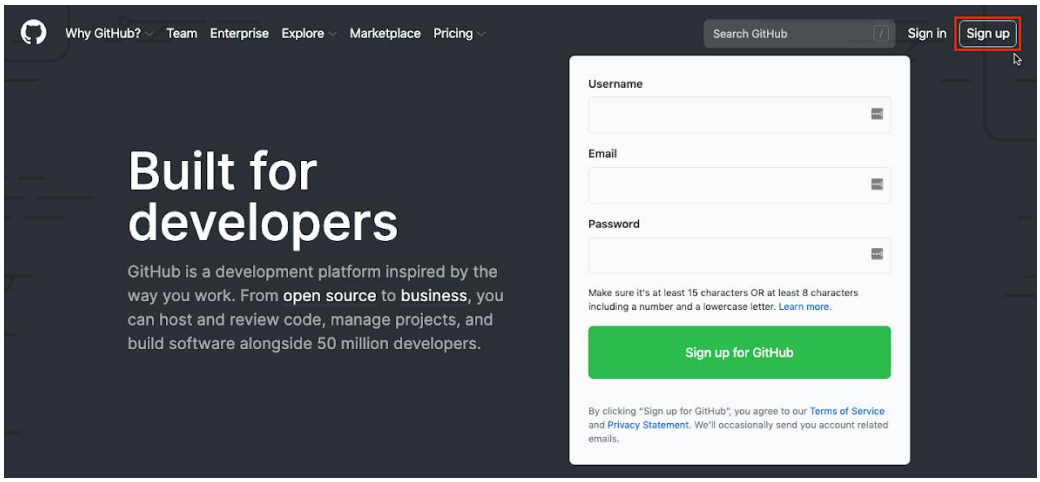
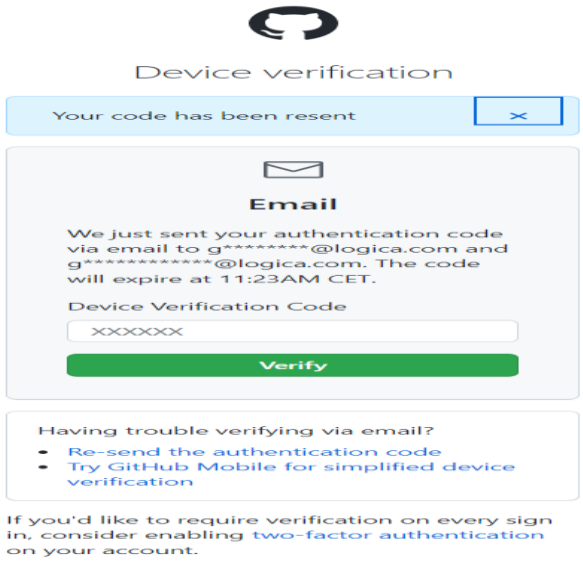
EXPERIMENT 18: Register on GitHub deploy a sample website

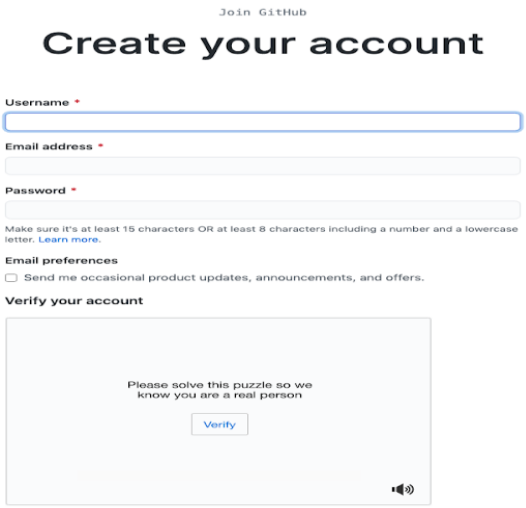
1. **Creating a GitHub Account**



2.



3.



* Click on Sign Up

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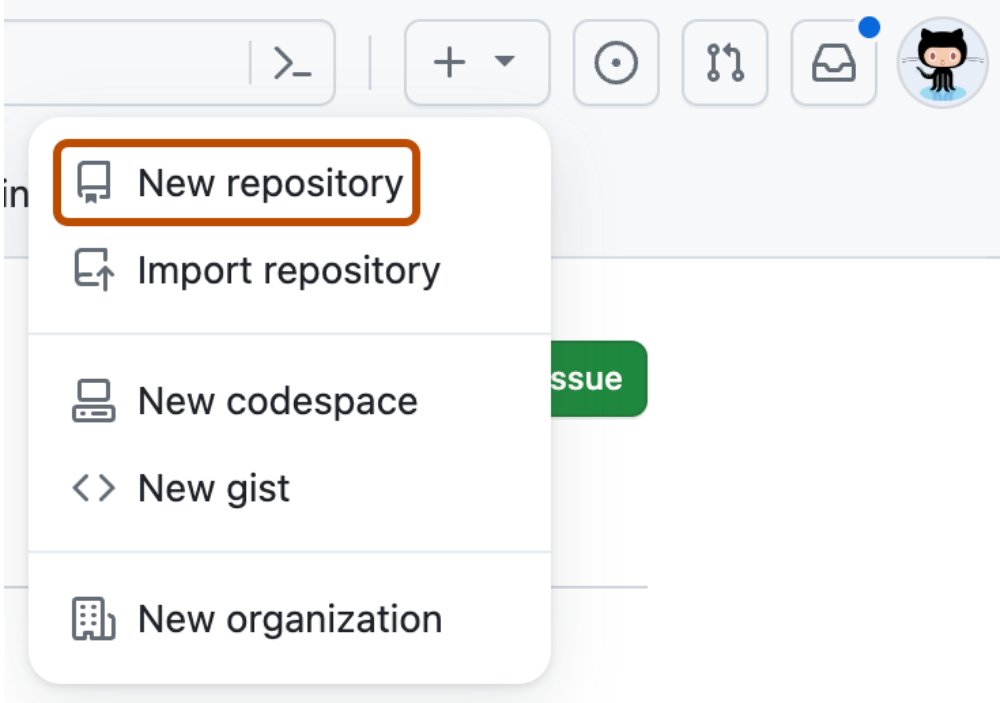
* Enter email, username, and password
* Verify your email address
* Choose the free plan
* Account is created successfully

Creating a Repository :



Steps:

1. Login to GitHub
2. Click on "+" → New Repository
3. Enter repository name (e.g., *myproject*)
4. Add description (optional)
5. Select Public or Private
6. Click Create Repository

Deploying the Project :

Upload Project to GitHub

Go to GitHub

```
git init
git add .
```

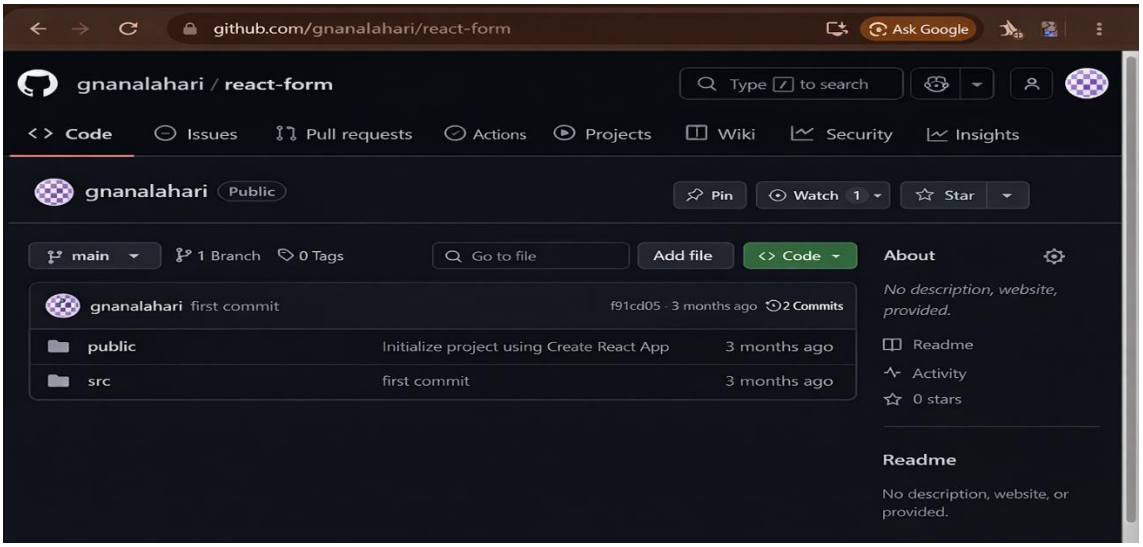
REDGNO : 24B01A1226
git commit -m "django project"
git remote add origin https://github.com/Mounishu/myproject.git
git push -u origin main

Deploy Using Render

Go to 👉 Render

Steps:

1. Login using GitHub
2. Click **New** → **Web Service**
3. Select your repository



EXPERIMENT 19 : create a webpage in Django using bootstarp for a menubar to a college which have separate welcome pages to each branch

CODE :

settings.py:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
```

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```
    'django.contrib.staticfiles',
    'collegeapp',
]
```

collegeapp/views.py :

```
from django.shortcuts import render
def home(request):
    return render(request, 'home.html')
def branch(request, name):
    return render(request, 'branch.html', {'branch': name})
```

collegeapp/urls.py :

```
from django.urls import path
from . import views
urlpatterns = [
    path('', views.home, name='home'),
    path('branch/<str:name>/', views.branch, name='branch'),
]
```

collegeproject/urls.py :

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('collegeapp.urls')),
]
```

collegeapp/templates/base.htm :

```
<!DOCTYPE html>
<html>
<head>
    <title>College Website</title>
```

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```
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body>

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">

  <div class="container-fluid">

    <a class="navbar-brand" href="{% url 'home' %}">My College</a>

    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target="#nav">

      <span class="navbar-toggler-icon"></span>

    </button>

    <div class="collapse navbar-collapse" id="nav">

      <ul class="navbar-nav">

        <!-- Home -->

        <li class="nav-item">

          <a class="nav-link" href="{% url 'home' %}">Home</a

        <!-- Dropdown -->

        <li class="nav-item dropdown">

          <a class="nav-link dropdown-toggle" href="#" data-bs-
toggle="dropdown">

            Branches

          </a>

          <ul class="dropdown-menu">

            <li><a class="dropdown-item" href="{% url 'branch' 'cse'
%}">CSE</a></li>

            <li><a class="dropdown-item" href="{% url 'branch' 'it' %}">IT</a></li>

            <li><a class="dropdown-item" href="{% url 'branch' 'civil'
%}">Civil</a></li>

            <li><a class="dropdown-item" href="{% url 'branch' 'mech'
%}">Mech</a></li>
```

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```
                </ul>

              </li>

            </ul>

          </div></div>

        </nav>

        <!-- Content -->

        <div class="container mt-4">

          {% block content %}

          {% endblock %}

        </div>

        <script
        src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.
        js"></script>

        </body>

        </html>
```

home.html :

```
{% extends 'base.html' %}

<h1>Welcome to My College</h1>

<p>Select a branch from the dropdown menu.</p>

{% endblock %}
```

branch.html :

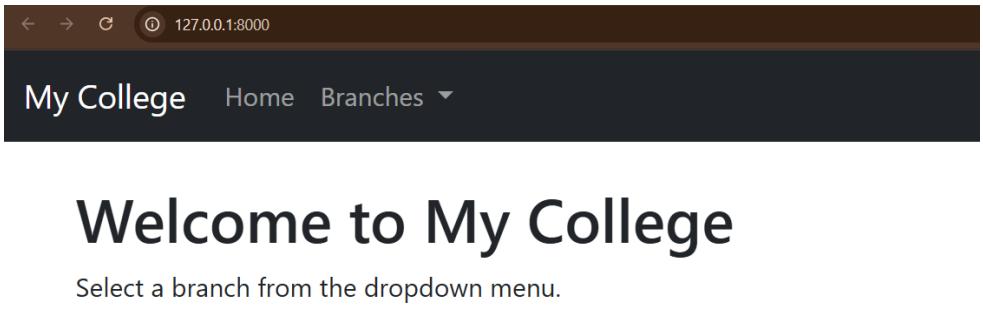
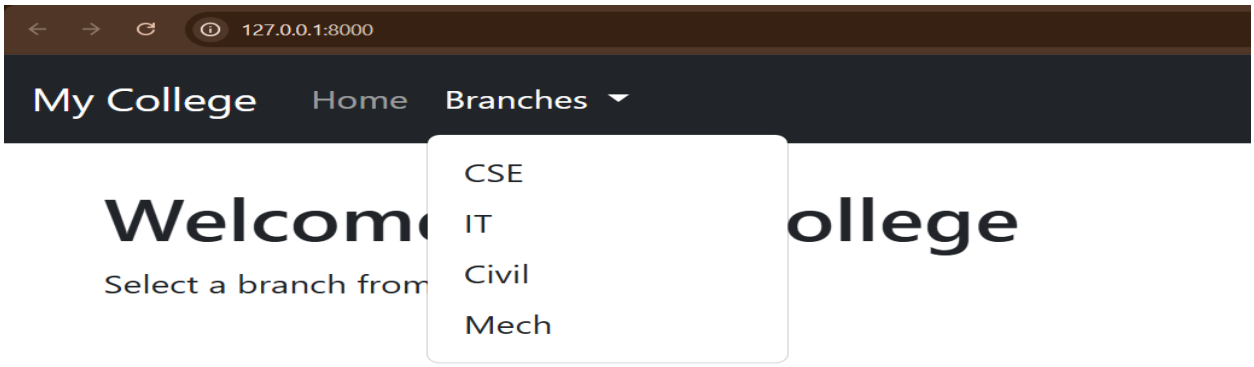
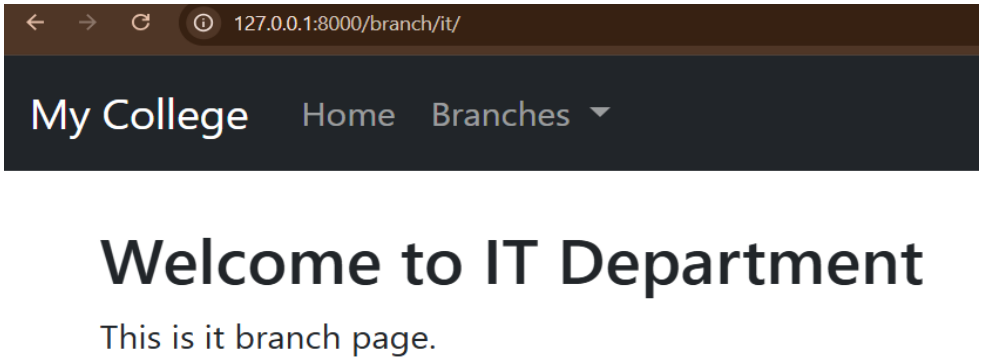
```
{% extends 'base.html' %}

{% block content %}

<h2>Welcome to {{ branch|upper }} Department</h2>

<p>This is {{ branch }} branch page.</p>

{% endblock %}
```

OUTPUT :

REDGNO : 24B01A1226